MyLearna
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MATHEMATICS

Interpret proportional information in data and real decisions

Name: ____________________

Date: ____________________

Step 12

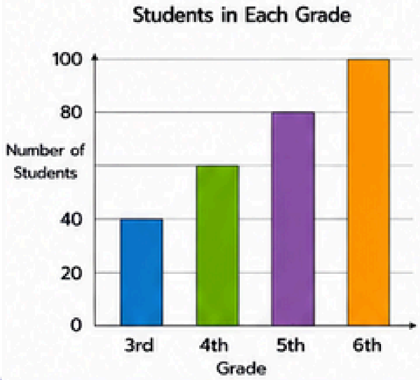
Upper Elementary

Look carefully. Think about parts of a whole.
Show your work by colouring, drawing, or writing.

Today I am learning to interpret proportional information in data and real decisions.

1. Read the Bar Graph

The bar graph shows the number of students in each grade.



Answer the questions.

a) What is the total number of students? ____________

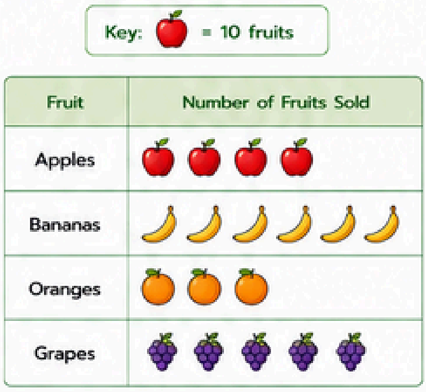
b) What percent of students are in 5th grade? ____________ %

c) What fraction of students are in 4th grade? ____________

d) How many more students are in 6th grade than 3rd grade? ____________

2. Pictograph

The pictograph shows the fruits sold in one day at a fruit stand.



Answer the questions.

a) Which fruit was sold the most? ____________

b) How many bananas were sold? ____________

c) What percent of the fruits sold were oranges? ____________ %

d) If 30 more apples were sold, how many apples would be sold in total? ____________

3. Table of Data

The table shows how long it takes to walk different distances.

Distance (km)	Time (hours)	Time per km (hours)
2	1	______
4	2	______
6	3	______
8	4	______
10	5	______

Answer the questions.

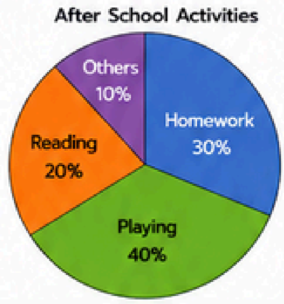
a) What is the unit rate (time per km)? ____________

b) How long will it take to walk 15 km at the same rate? ____________ hours

c) If the walk takes 7 hours at the same rate, how far will the person walk? ____________ km

4. Circle Graph

The circle graph shows how a student spends 60 minutes after school.



Answer the questions.

a) How many minutes are spent on playing? ____________ minutes

b) How many minutes are spent on reading? ____________ minutes

c) If homework time increases to 40%, how many minutes will be spent on homework? ____________ minutes

5. Comparing Proportions

Use the table to compare and answer.
A recipe uses 3 cups of flour to make 18 cookies.

Recipe	Cups of Flour	Number of Cookies	Cookies per Cup
Recipe A	3	18	______
Recipe B	2	12	______
Recipe C	4	32	______

Answer the questions.

a) Which recipe has the greatest number of cookies per cup of flour?

b) If you have 5 cups of flour, which recipe will make the most cookies?

c) How many cookies will Recipe B make if you use 5 cups of flour? ____________ cookies

6. Financial Decisions

A television costs $600.
Use the table to answer the questions.

Payment Option	Percent of Price	Amount to Pay	You Save
Pay in full	100%	$ ______	$ 0
10% discount	90%	$ ______	$ ______
20% discount	80%	$ ______	$ ______
30% discount	70%	$ ______	$ ______

Answer the questions.

a) How much will you pay with a 20% discount? $ ____________

b) How much will you save with a 30% discount? $ ____________

c) Which payment option is the best deal? Why? ____________________
____

7. Real-Life Problem Solving

Solve each problem.

a) A car travels 240 km using 12 litres of fuel. At the same rate, how far can it travel using 20 litres of fuel?

Answer: ____________ km

b) A store sells 6 notebooks for $15. At the same price, how much will 10 notebooks cost?

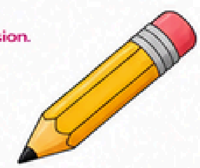
Answer: $ ____________

c) A class has 24 students. 18 students like soccer. What percent of the class likes soccer?

Answer: ____________ %

8. Think and Decide

Use proportional information to make the best decision.



You have $50 to spend on pencils.

Store A: 5 pencils for $6
Store B: 8 pencils for $9
Store C: 12 pencils for $13

a) Which store gives you the best value (lowest price per pencil)? ____________

b) How many pencils can you get from that store with $50? ____________ pencils

c) Explain how you decided.
____
____

I can...

- ☐ read and interpret data displays.
- ☐ find and compare unit rates.
- ☐ use proportional thinking to solve real problems.
- ☐ make informed decisions using data.

My Learning Goal

I will use proportional information to understand data and make good decisions.
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